THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
NMTC JUNIOR LEVEL – IX & X Grades FINAL

NMTC – 2023

Instructions:

1. **Answer all questions. Each question carries 10 marks.**
2. **Elegant and innovative solutions will get extra marks.**
3. **Diagram and Justification should be given wherever necessary.**
4. **Before start answering, fill in the FACE SLIP completely.**
5. **Your 'rough work" should be in the answer sheet itself.**
6. **Maximum Time allowed is THREE hours.**

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1. *x, y, z* are positive reals and $(x + y + z)^3 = 32xyz$. Find the numerical limits between which the expression $\frac{x^4+y^4+z^4}{(x+y+z)^4}$ lies?
2. $A_1A_2A_3A_4A_5A_6A_7$ is a regular heptagon. Prove that $\frac{{A_1A_4}^3}{{A_1A_2}^3} - \frac{A_1A_7+2A_1A_6}{A_1A_5-A_1A_3} = 1$
3. ABCD is a square whose side is 1 unit. Let n be ab arbitrary natural number. A figure is drawn inside the square consisting of only line segments, having a total length greater than 2n. (This figure can have many pieces of single line segments intersecting or non intersecting). Prove that for some straight line L which is parallel in a side of the square must cross the figure at least (n+1) times.
4. *m* is a natural number. If (*2m+1*) and (*3m+1*) are perfect squares. Then prove that m is divisible by 40.
5. Given 69 distinct positive integers not exceeding 100, prove that one can choose four of them *a, b, c, d* such that *a* < *b* < *c* and *a* + *b* + *c* = *d*. Is this statement true for 68?
6. *m, n* are integers such that $n^2(m^2+1) + m^2(n^2+16) = 448$. Then find all possible ordered pairs (*m, n*).
7. BD is the bisector of ∠ABC of triangle ABC. The circumcircle of triangle BCD and ABD cut AB and BC at E and F respectively. Show that AE = CF.
8. *a* is a two-digit number, *b* is a three-digit number. *a* increased by *b* percent is equals to the *b* decreased by *a* percent. Find all possible ordered pairs (*a, b*).

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